# UNIT - II

## Analysis of Indeterminate Structures

## 1. Introduction

- A structure is **statically determinate** if all reactions and internal forces can be found using only the **equations of static equilibrium** .
- A structure is **statically indeterminate (redundant)** if the equilibrium equations are **not sufficient** ; extra equations from **compatibility of deformation** (geometry) are needed.
- **Equilibrium equations (plane structures):** $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M = 0$ — three equations.

| Basis | Determinate | Indeterminate |
|---|---|---|
| Analysis | equilibrium equations only | equilibrium + compatibility (material property EI, AE needed) |
| Forces depend on | loads and geometry only | also on stiffness (EI, AE) of members |
| Temperature / settlement / lack of fit | no stresses developed | stresses are developed |
| Failure | local failure → collapse | redistribution of moments; more safety |
| Material / sections | larger sections needed | economical, smaller sections |
| Deflection | larger | smaller (more stiff) |
| Examples | S.S. beam, cantilever, 3-hinged arch | fixed beam, continuous beam, propped cantilever, fixed portal frame |

### Advantages of indeterminate structures

- Smaller bending moments, so **economical sections** ; **stiffer** with small deflections.
- **Redundancy gives safety** — load redistributes if one member fails.
- Better stability and **more aesthetic** (slender) designs; used in bridges and multistorey buildings.

### Disadvantages

- Complex analysis; sensitive to **support settlement, temperature, lack of fit** (they induce stresses).
- Skilled design and construction needed; connections are costly.

## 2. Static Indeterminacy ($D_s$)

- **Degree of static indeterminacy = number of unknown forces – number of equilibrium equations** = number of **redundants** (extra reactions or members that can be removed without making the structure unstable).
- **External indeterminacy ($D_{se}$):** reactions exceed the available equilibrium equations.
- **Internal indeterminacy ($D_{si}$):** member forces cannot be found even if all reactions are known.
- **Total $D_s = D_{se} + D_{si}$.**

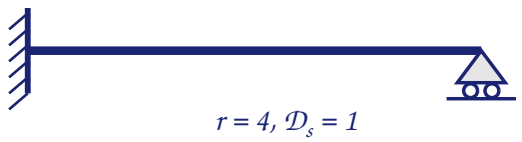
### Formulae

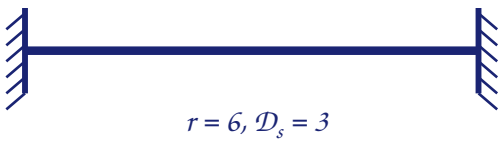
| Structure | Degree of static indeterminacy |
|---|---|
| Plane truss (pin-jointed) | $D_s = m + r - 2j$ ($m$ = members, $r$ = reaction components, $j$ = joints) |
| Plane truss, external | $D_{se} = r - 3$ |
| Plane truss, internal | $D_{si} = m - (2j - 3)$ |
| Plane rigid frame | $D_s = 3m + r - 3j$ (– number of released conditions c) |
| Beam (no hinges) | $D_s = r - 3$ |
| Beam with internal hinges | $D_s = r - 3 - e$ ($e$ = extra condition equations, e.g. hinge: 1) |
| Space truss | $D_s = m + r - 3j$ |
| Space frame | $D_s = 6m + r - 6j$ |

- For **closed loops** (rings): each closed box adds **3** degrees of indeterminacy.
- For **beams under vertical loads only**, the horizontal reaction is zero so one equation less is needed: a fixed beam is then taken as **2** degrees.

→ Stability check: reactions must not be all parallel or concurrent; $D_s$ negative means the structure is unstable (mechanism).

### Examples of static indeterminacy




**Propped cantilever ($D_s = 1$) and Fixed beam ($D_s = 3$)**

| Structure | Calculation | $D_s$ |
|---|---|---|
| Simply supported beam | $r = 3$; $3 - 3$ | 0 |
| Cantilever | $r = 3$; $3 - 3$ | 0 |
| Propped cantilever | fixed (3) + roller (1) = 4; $4 - 3$ | 1 |

| Structure | Calculation | $D_s$ |
|---|---|---|
| Fixed beam | 3 + 3 = 6; 6 − 3 | 3 (2 for vertical loads) |
| Two-span continuous beam (pin + 2 rollers) | 2 + 1 + 1 = 4; 4 − 3 | 1 |
| Three-span continuous beam (pin + 3 rollers) | 5 − 3 | 2 |
| Portal frame, hinged at both bases | 3m + r − 3j = 9 + 4 − 12 | 1 |
| Portal frame, fixed at both bases | 9 + 6 − 12 | 3 |
| Three-hinged arch/frame | r = 4, e = 1: 4 − 3 − 1 | 0 |
| Plane truss with 6 members, 4 joints, r = 3 | 6 + 3 − 8 | 1 (internal) |

- For continuous beam with n supports (one pin, rest rollers): $D_s = n - 2$ .

## 3. Kinematic Indeterminacy ($D_k$) — Degrees of Freedom

- **Degree of kinematic indeterminacy** = number of **independent joint displacements (translations and rotations)** that must be known to describe the deformed shape. It is the number of unknowns in the **displacement (stiffness) method** .
- A **fixed support** has no displacement; a **pin** can rotate; a **free end** can translate and rotate.

### Formulae

| Structure | $D_k$ |
|---|---|
| Plane truss | $D_k = 2j - r$ (each free joint has 2 translations) |
| Plane frame (axial deformation considered) | $D_k = 3j - r$ |
| Plane frame, axially rigid members | $D_k = 3j - r - m$ |
| Continuous beam (axial effect neglected) | $D_k = 2j - r$ or number of joint rotations + translations |
| Space truss | $D_k = 3j - r$ |
| Space frame | $D_k = 6j - r$ |

| Structure | Free displacements | $D_k$ | $D_s$ |
|---|---|---|---|
| Fixed beam | none | 0 | 3 |
| Propped cantilever | rotation at the pinned/roller end | 1 | 1 |

| Structure | Free displacements | $D_k$ | $D_s$ |
|---|---|---|---|
| Cantilever | rotation + deflection at free end (axial neglected) | 2 | 0 |
| Two-span continuous beam, simple supports | rotation at A, B, C | 3 | 1 |
| Two-span continuous beam, A fixed | rotation at B, C | 2 | 2 |
| Portal frame fixed bases (axial neglected) | rotation at B, C + sway | 3 | 3 |
| Portal frame hinged bases (axial neglected) | rotation at A, B, C, D + sway | 5 | 1 |
| Portal frame fixed bases (axial included) | u, v, θ at B and C | 6 | 3 |

→ Rule: Moment distribution/slope-deflection need $D_k$ unknowns; force methods (Castigliano, flexibility) need $D_s$ unknowns. Choose the method with fewer unknowns.

### Static vs Kinematic indeterminacy

| Basis | Static | Kinematic |
|---|---|---|
| Unknowns | redundant forces | joint displacements |
| Method used | force / flexibility | displacement / stiffness |
| Examples of methods | Castigliano II, consistent deformation, three-moment | slope-deflection, moment distribution, stiffness matrix |
| Fixed beam | 3 (or 2) | 0 |
| Cantilever | 0 | 2 |
| Equations | compatibility | equilibrium of joints |

## 4. Methods of Analysis of Indeterminate Structures

**Force (flexibility) methods**
Castigliano II, consistent deformation, theorem of three moments, column analogy, flexibility matrix

**Displacement (stiffness) methods**
slope-deflection, moment distribution, Kani's method, stiffness matrix

**Approximate methods**
portal method, cantilever method (for lateral loads)

**Energy and others**
unit load method, virtual work, finite element

- **Common idea:** choose the redundants → make the structure determinate (**released / primary**

**structure** ) → find displacement of the primary structure at the redundant locations (due to loads and due to unit redundants) → impose compatibility conditions.

## 5. Castigliano's Second Theorem (Theorem of Least Work)

**Statement**

**"For a statically indeterminate structure, the partial derivative of the total strain energy with respect to each redundant force (reaction or internal force) is zero, provided the supports are unyielding and no temperature or lack-of-fit effects are present."**

$\partial U / \partial R = 0$ ($R$ = redundant)

- It means: **of all the stress distributions that satisfy equilibrium, the actual one is that which makes the strain energy minimum** — hence "least work".

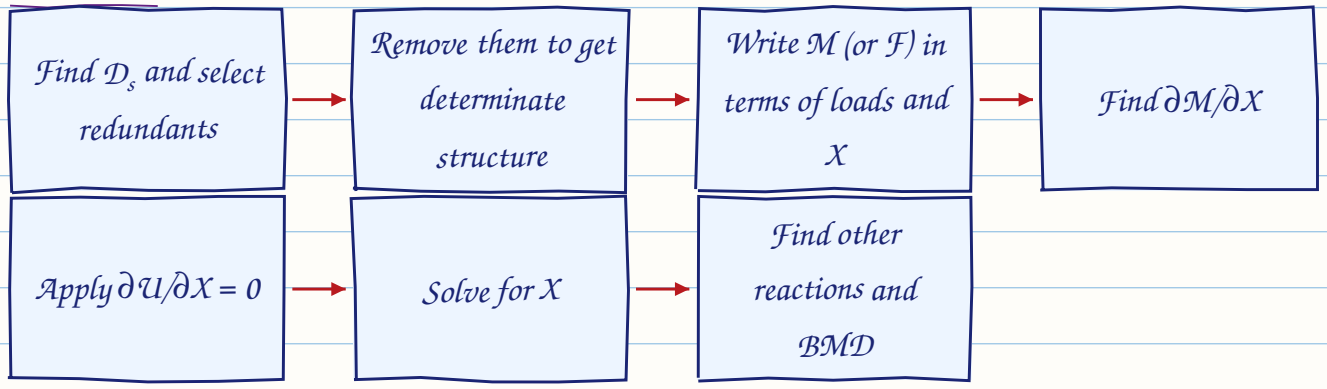
### Proof (simple)

- From Castigliano's first theorem the deflection at the point of the redundant reaction $R$ is $\delta = \partial U/\partial R$.
- Because the support is unyielding, the deflection in the direction of $R$ is **zero** ; hence $\partial U/\partial R = 0$ .
- If the support settles by $\Delta$ in the direction of $R$, then $\partial U/\partial R = \Delta$ (equal to the settlement). Temperature change and lack of fit add extra terms (see section 9).

### Equations

- For a beam: $\partial U/\partial R = \int (M/EI)(\partial M/\partial R)\,dx = 0$ .
- For a truss: $\partial U/\partial X = \Sigma (F L/AE)(\partial F/\partial X) = 0$ with $F = F_0 + kX$.
- With **n** redundants $X_1 \ldots X_n$: write n equations $\partial U/\partial X_i = 0$ → solve simultaneously.

### Procedure

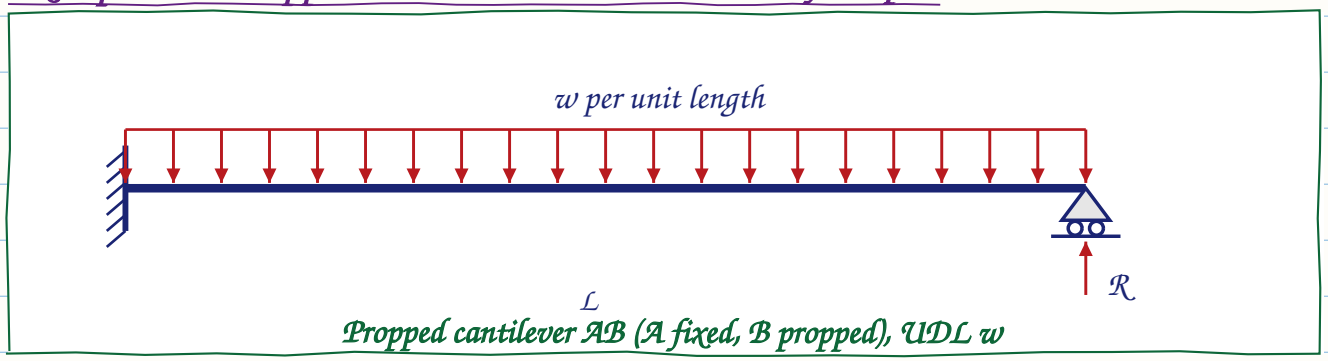



1. Determine the degree of indeterminacy and select redundant(s) — pick those that make the work easy (e.g., a prop reaction).
2. Express reactions, $M$ or $F$ in terms of the **external loads and redundants** .
3. Form $U$ and differentiate: $\partial U/\partial X = 0$ (differentiate first, integrate later).
4. Solve; then find the remaining reactions by equilibrium, then SFD and BMD.

→ Not valid if supports sink or if there is temperature change / lack of fit unless the extra terms are included.

## 6. Indeterminate Beams by Castigliano II — Examples

### Example 1 — Propped cantilever with UDL w over full span




Propped cantilever AB (A fixed, B propped), UDL w

- Take $R_B = R$ as redundant. Measure $x$ from B. $M = Rx - w x^2/2$; $\partial M/\partial R = x$.
- $\partial U/\partial R = (1/EI)\int_0^L (Rx - w x^2/2)\, x\, dx = 0 \rightarrow R L^3/3 - w L^4/8 = 0$.

**Result**

$R_B = 3wL/8$, $R_A = 5wL/8$, $M_A = -wL^2/8$ (hogging), max sagging $M = 9wL^2/128$ at $x = 3L/8$ from B.

### Example 2 — Propped cantilever with a central point load P

- $x$ from B: for $0 \le x \le L/2$, $M = Rx$; for $L/2 \le x \le L$, $M = Rx - P(x - L/2)$; $\partial M/\partial R = x$.
- $\int M x\, dx = R L^3/3 - P\int_{L/2}^{L} (x - L/2)\, x\, dx = R L^3/3 - 5PL^3/48 = 0$.

**Result**

$R_B = 5P/16$, $R_A = 11P/16$, $M_A = -3PL/16$, $M_C$ (under load) $= +5PL/32$.

### Example 3 — Two-span continuous beam, equal spans L, UDL w on both spans

- Support B (middle) is taken as redundant: $R_B = R$; by symmetry $R_A = R_C = wL - R/2$.
- Span AB ($x$ from A): $M = (wL - R/2)x - w x^2/2$; $\partial M/\partial R = -x/2$.
- $\partial U/\partial R = 2\int_0^L M(-x/2)\, dx/EI = 0 \rightarrow (wL - R/2)L^3/3 = wL^4/8$.

**Result**

$R_B = 5wL/4$, $R_A = R_C = 3wL/8$, $M_B = -wL^2/8$, max sagging $= 9wL^2/128$ at $3L/8$ from the end supports.

- Check: $R_A + R_B + R_C = 3wL/8 + 5wL/4 + 3wL/8 = 2wL$ ✓ (total load).

### Example 4 — Numerical

**Problem**

A propped cantilever 6 m long carries a UDL of 20 kN/m over the entire span. Find the prop reaction and fixed-end moment.

- $R_B = 3wL/8 = 3 \times 20 \times 6/8 = 45$ kN; $R_A = 120 - 45 = 75$ kN.

- $M_A = -wL^2/8 = -20 \times 36 / 8 = \mathbf{-90\ kN{\cdot}m}$ .

### Effect of settlement of the prop (support B sinks by Δ)

- Compatibility: deflection of the cantilever at $B = wL^4/8EI - RL^3/3EI = \Delta$ (settlement).
- Hence $\mathbf{R_B = 3wL/8 - 3EI\Delta/L^3}$ — settlement of the prop **reduces** the prop reaction.

## 7. Indeterminate Pin-Jointed Trusses

### Types of redundancy in trusses

- **External redundancy:** extra support reactions ($r > 3$). Redundant = a reaction. Eg: hinged supports at both ends ($r = 4$).
- **Internal redundancy:** more members than needed ($m > 2j - 3$). Redundant = force in a **member** . Eg: a panel with both diagonals.
- **Degree = $(m + r) - 2j$** . In this unit, up to **two** redundants (internal and/or external).

### Method (Castigliano II for trusses)

- Select the redundant(s) $X_1$ (and $X_2$). For internal redundants **cut the member** and apply $X$ on both sides of the cut. For external redundants remove the support.
- Member forces: $\mathbf{F = F_0 + k_1 X_1 + k_2 X_2}$
- $F_0$ = force in the **primary (determinate) truss** due to actual loads. $k_1$, $k_2$ = force due to a **unit** value of $X_1$, $X_2$ (the cut member itself has $k = 1$).

**Two redundants (simultaneous equations)**

$$\partial U/\partial X = \Sigma\,(F_0 + kX)\,kL/AE = 0 \Rightarrow X = -\Sigma\,F_0\,kL/AE \div \Sigma\,k^2 L/AE$$

$$X_1 \Sigma\,k_1^2 L/AE + X_2 \Sigma\,k_1 k_2 L/AE = -\Sigma\,F_0 k_1 L/AE$$

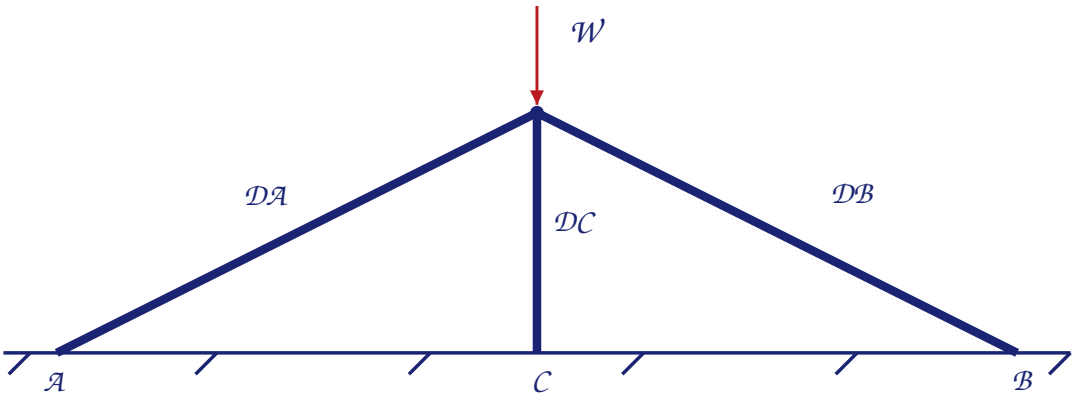
$$X_1 \Sigma\,k_1 k_2 L/AE + X_2 \Sigma\,k_2^2 L/AE = -\Sigma\,F_0 k_2 L/AE$$

- If $AE$ is same for all members it cancels out. The redundant member itself is also included in the summations.
- **Lack of fit λ** in a redundant member and **temperature change** appear as: $\Sigma\,(F_0 + kX)\,kL/AE = -\Sigma\,k\lambda$ ($\lambda = \alpha T L$ or the lack of fit; sign as per convention).

Select redundants → Find $F_0$ (loads on primary truss) → Find $k_1$, $k_2$ (unit redundants) → Tabulate $\Sigma\,F_0\,kL$, $\Sigma\,k^2 L$

Solve for $X$ → Final $F = F_0 + kX$

### Example 5 — Three-bar concurrent truss (one degree, external/internal)




Three bars DA, DC, DB meeting at D; load W at D

- Vertical bar DC length $L$; inclined bars DA and DB at angle $\theta$ to the vertical, length $L/\cos\theta$. All have the same $AE$.
- Redundant: $X$ = force in DC. Equilibrium at D: $2F\cos\theta + X = W \rightarrow$ **$F = (W - X)/2\cos\theta$** ($F$ = force in each inclined bar).
- $U = X^2 L / 2AE + 2F^2 (L/\cos\theta)/2AE$; $\partial F/\partial X = -1/(2\cos\theta)$.
- $\partial U/\partial X = X L/AE + 2F(-1/2\cos\theta)(L/\cos\theta)/AE = 0 \rightarrow X = F/\cos^2\theta = (W - X)/(2\cos^3\theta)$.

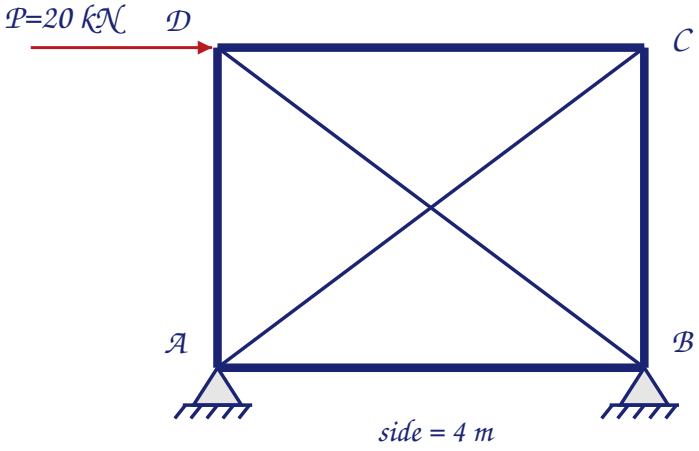
**Result**

**$X = W/(1 + 2\cos^3\theta)$** (force in the vertical bar, tension)

- Numerical: $\theta = 30°$, $W = 100$ kN $\rightarrow \cos^3 30° = 0.6495 \rightarrow$ **$X = 100/2.299 = 43.5$ kN**; $F = (100 - 43.5)/(2 \times 0.866) =$ **32.6 kN** in each inclined bar.
- Check: $2 \times 32.6 \times 0.866 + 43.5 = 100$ ✓.

## 8. Solution of Trusses with Two Redundants

### Example 6 — Square panel truss (side 4 m) with both diagonals, hinged at A and B




$m = 6, r = 4, j = 4$

$D_s = 6+4-8 = 2$

Square panel ABCD: both diagonals; pinned at A and B; load P = 20 kN at D

- Redundants: $X_1$ = horizontal reaction at B (external), $X_2$ = force in diagonal AC (internal, cut). Same $AE$ for all members.
- Primary truss: A hinged, B roller (vertical only), AC cut.

| Member | L (m) | $F_0$ (kN) | $k_1$ | $k_2$ |
|---|---|---|---|---|

| Member | $L$ (m) | $F_0$ (kN) | $k_1$ | $k_2$ |
|---|---|---|---|---|
| AB | 4 | +20 | +1 | −0.707 |
| BC | 4 | 0 | 0 | −0.707 |
| CD | 4 | 0 | 0 | −0.707 |
| DA | 4 | +20 | 0 | −0.707 |
| AC | 5.657 | 0 | 0 | +1 |
| BD | 5.657 | −28.28 | 0 | +1 |

- Sums: $\Sigma k_1^2 L = 4$; $\Sigma k_1 k_2 L = -2.83$; $\Sigma k_2^2 L = 19.31$; $\Sigma F_0 k_1 L = 80$; $\Sigma F_0 k_2 L = -273.14$.
- Equations ($AE$ cancels): $4 X_1 - 2.83 X_2 = -80$ and $-2.83 X_1 + 19.31 X_2 = +273.14$.
- Solving: $X_1 = -11.16$ kN ($H_B$ acts opposite to the assumed direction), $X_2 = +12.51$ kN (tension in AC).

| Member | Final force $F = F_0 + k_1 X_1 + k_2 X_2$ | Nature |
|---|---|---|
| AB | $20 - 11.16 - 8.85 = 0$ | zero force |
| BC | $-8.85$ | compression |
| CD | $-8.85$ | compression |
| DA | $20 - 8.85 = +11.16$ | tension |
| AC | $+12.51$ | tension |
| BD | $-28.28 + 12.51 = -15.78$ | compression |

- **Check joint D:** horizontal: $20 - 8.85 - 15.78\cos45° = 0$ ✓; vertical: $-11.16 + 15.78\sin45° = 0$ ✓.

> **Tip**
> Choose the redundants so that the primary truss is simple and as many $k$ values as possible are zero — it reduces the sums.

## 9. Indeterminate Trusses — Special Effects

- **Temperature rise $\Delta T$** in members: $\lambda = \alpha \Delta T L$; in a redundant truss it induces forces: $\Sigma (F_0 + k X) k L/AE + \Sigma k \alpha \Delta T L = 0$.
- **Lack of fit** (member too short/long by $e$): treat as an initial elongation $e$; the same type of term.
- **Support settlement $\delta_s$** at the redundant: $\partial U/\partial X = -\delta_s$ (sign by direction).
- **Self-stressing** truss: stresses appear even without external loads ($F_0 = 0$).
- **Effect of redundant member:** total strain energy is minimum; deflection is reduced.

## 10. Comparison — Castigliano I vs Castigliano II

| Basis | Castigliano's first theorem | Castigliano's second theorem |
|---|---|---|
| Statement | $\partial U/\partial P = \delta$ (deflection at $P$) | $\partial U/\partial R = 0$ (redundant $R$, unyielding support) |
| Used to find | deflection / slope | redundant reactions or member forces |
| Structure | determinate (or indeterminate once redundants are known) | indeterminate |
| Dummy load | needed if no load at the point | not needed; redundant is the variable |
| Final step | set dummy load to zero | solve the equation for $X$ |

## 11. Additional Solved Illustrations

### Illustration 1 — Propped cantilever with a point load at the middle (numerical)

- $L = 8$ m, $P = 48$ kN at the middle: $R_B = 5P/16 =$ **15 kN**; $R_A = 33$ kN; $M_A = -3PL/16 =$ **– 72 kN·m**; $M_C = 5PL/32 =$ **+ 60 kN·m**.

### Illustration 2 — Two-span beam, numerical

- Two equal spans of 5 m, UDL 24 kN/m: $R_B = 5wL/4 =$ **150 kN**; $R_A = R_C = 3wL/8 =$ **45 kN**; $M_B = -wL^2/8 =$ **– 75 kN·m**.

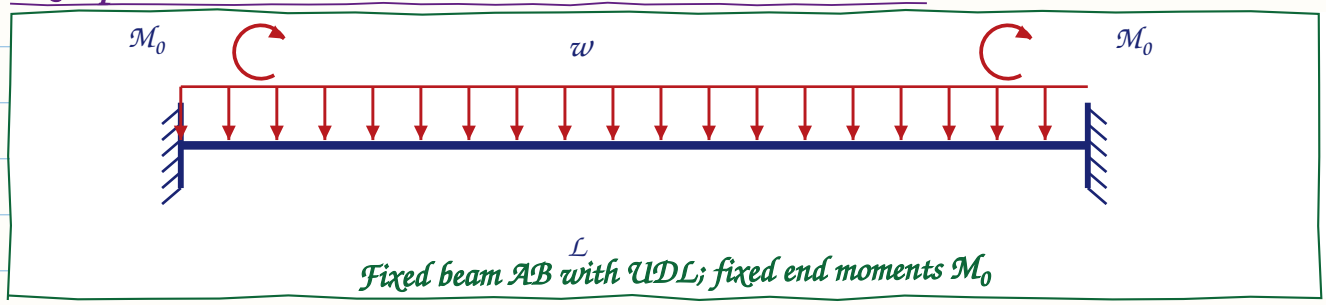
### Illustration 3 — Bars with a common joint (check)

- If $\theta = 0$, the three bars are all vertical: $X = W / 3$ — each carries equal load, as expected for equal $AE$.

## 11A. Fixed Beams by Castigliano II (link to Unit III)

- A fixed beam has $D_s = 3$; for vertical loads the horizontal reaction is zero, so take the **end moments** as redundants. By symmetry (symmetric loading) $M_A = M_B = M_0$.

### Example 7 — Fixed beam with UDL w over the whole span




Fixed beam AB with UDL; fixed end moments $M_0$

- Reactions $R_A = R_B = wL/2$. Taking sagging as positive: $M = wLx/2 - w x^2/2 - M_0$; $\partial M/\partial M_0 = -1$.
- $\partial U/\partial M_0 = 0$ (no rotation at the fixed end) $\rightarrow \int_0^L (wLx/2 - w x^2/2 - M_0)\, dx = 0$
- $\rightarrow w L^3/4 - w L^3/6 - M_0 L = 0 \rightarrow$ **$M_0 = w L^2 / 12$**.
- Centre moment = $wL^2/8 - wL^2/12 =$ **$+ wL^2/24$** (sagging). Points of contraflexure at 0.211 $L$

from each end.

### Example 8 — Fixed beam with central point load P

- For $0 \le x \le L/2$: $M = P x/2 - M_0$. By symmetry integrate over half: $\int_0^{L/2} (P x/2 - M_0)\, dx = 0$
- $\rightarrow P L^2/16 - M_0 L/2 = 0 \rightarrow \mathbf{M_0 = P L / 8}$ ; centre moment $= PL/4 - PL/8 = \mathbf{+ PL/8}$ .
- These fixed-end moments are the basis of **slope-deflection and moment distribution** (Units IV and V).

## 11B. Cross-Check by Consistent Deformation (Compatibility Method)

- Same propped cantilever with UDL: remove the prop → free cantilever; deflection at B due to UDL = $w L^4 / 8EI$ (downward).
- Apply upward R at B → deflection = $R L^3 / 3EI$ (upward).
- Compatibility: $\mathbf{w L^4 / 8EI = R L^3 / 3EI \rightarrow R = 3wL/8}$ √ (same as Castigliano II).
- Two-span beam check: deflection at B of a simply supported beam of span 2L with UDL = 5 w $(2L)^4 / 384EI = 5wL^4/24EI$; due to R: $R(2L)^3/48EI = RL^3/6EI$; equate → $\mathbf{R = 5wL/4}$ √.

| Basis | Castigliano II | Consistent deformation |
|---|---|---|
| Idea | minimum strain energy | geometric compatibility at the redundant |
| Works with | M or F expressions in terms of X | standard deflection formulae |
| Best for | trusses, frames, curved members | beams with standard cases |
| Computation | integration needed | uses tabulated results — fast |

## 11C. Practice — Find $D_s$ and $D_k$ (with answers)

| No. | Structure | $D_s$ | $D_k$ |
|---|---|---|---|
| 1 | Beam fixed at A, rollers at B and C (two spans) | 5 − 3 = 2 | rotations at B, C = 2 |
| 2 | Plane truss: j = 6, m = 10, r = 3 | 10 + 3 − 12 = 1 | 2j − r = 9 |
| 3 | Fixed-fixed beam with an internal hinge | 6 − 3 − 1 = 2 | rotation on both sides of the hinge + deflection = 2 |
| 4 | Single-bay, two-storey frame, fixed bases (6 joints, 6 members, r = 6) | 18 + 6 − 18 = 6 | 4 rotations + 2 sways = 6 |
| 5 | Propped cantilever | 1 | 1 |
| 6 | Hinged-base portal frame | 1 | 5 |

| No. | Structure | $D_s$ | $D_k$ |
|---|---|---|---|
| 7 | Plane truss: $j = 7$, $m = 11$, $r = 3$ (simply supported) | $11 + 3 - 14 = 0$ (determinate) | $14 - 3 = 11$ |
| 8 | Closed rectangular ring (box) | 3 | – |

→ $D_k$ is different if axial deformation is considered or neglected — mention your assumption in the answer.

## 11D. Settlement, Temperature and Redundant Structures

- **Determinate structure:** no stresses due to settlement or temperature (it just moves).
- **Indeterminate structure:** settlement or temperature change **induces** stresses and reactions — a drawback, but also useful (prestressing by jacking supports).
- Fixed beam, one end sinks $\Delta$: end moments $\propto 6EI\Delta/L^2$ (see Unit III).
- A truss with a temperature rise in one member causes force in the redundant members.

## 11E. Stability, Mechanism and Geometry (supplement)

- Equation $D_s \geq 0$ is **necessary but not sufficient** . Check geometry: the reactions must not be concurrent or all parallel; no part should be free to move (mechanism).
- Example of unstable arrangement: a beam on three **roller** supports (all vertical) has $r = 3$ but cannot resist horizontal load → **unstable** .
- A truss with a **non-triangulated** panel is a mechanism (needs a diagonal).
- Check: **$D_s = 0$ and stable** → determinate; **$D_s > 0$ and stable** → indeterminate; **$D_s < 0$** → unstable.

## 12. Quick Revision — Unit II

### Standard results

Truss: $D_s = m + r - 2j$, $D_k = 2j - r$ | Frame: $D_s = 3m + r - 3j$, $D_k = 3j - r$ | Beam: $D_s = r - 3$ | Castigliano II: $\partial U/\partial R = 0$ | $F = F_0 + kX$ | $X = -\Sigma F_0 kL \div \Sigma k^2 L$.

Propped cantilever UDL: $R_B = 3wL/8$, $M_A = -wL^2/8$ | Point load at centre: $R_B = 5P/16$ | Two-span continuous beam UDL: $R_B = 5wL/4$.

### Possible exam questions

1. Differentiate between determinate and indeterminate structures. Explain static and kinematic indeterminacy.
2. Determine $D_s$ and $D_k$ for the given frames, beams and trusses.
3. State and prove Castigliano's second theorem (least work).
4. Analyse a propped cantilever/continuous beam using Castigliano's second theorem.

**5.** Analyse the indeterminate truss with one redundant member; two redundants (numerical).